

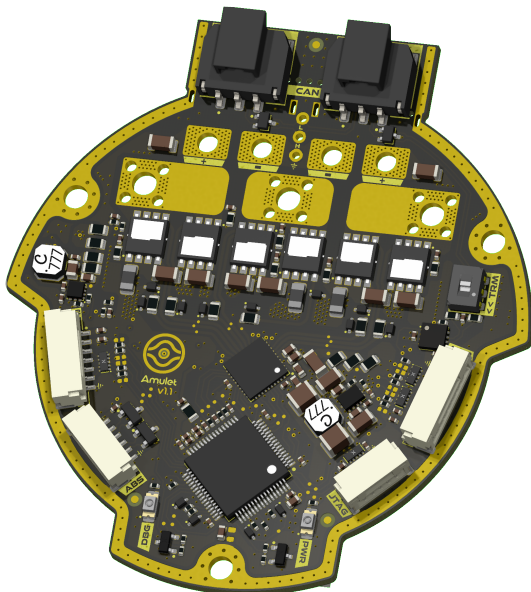
Amulet Motion Controller

Variant: NO_XT60

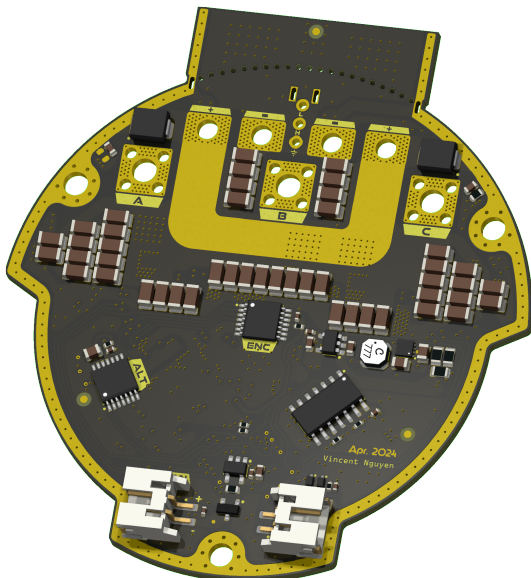
2024-11-21
Rev 1.1

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TOP VIEW



BOTTOM VIEW



DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

NOTES

Schematic based off Josh Pieper's moteus controllers.

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

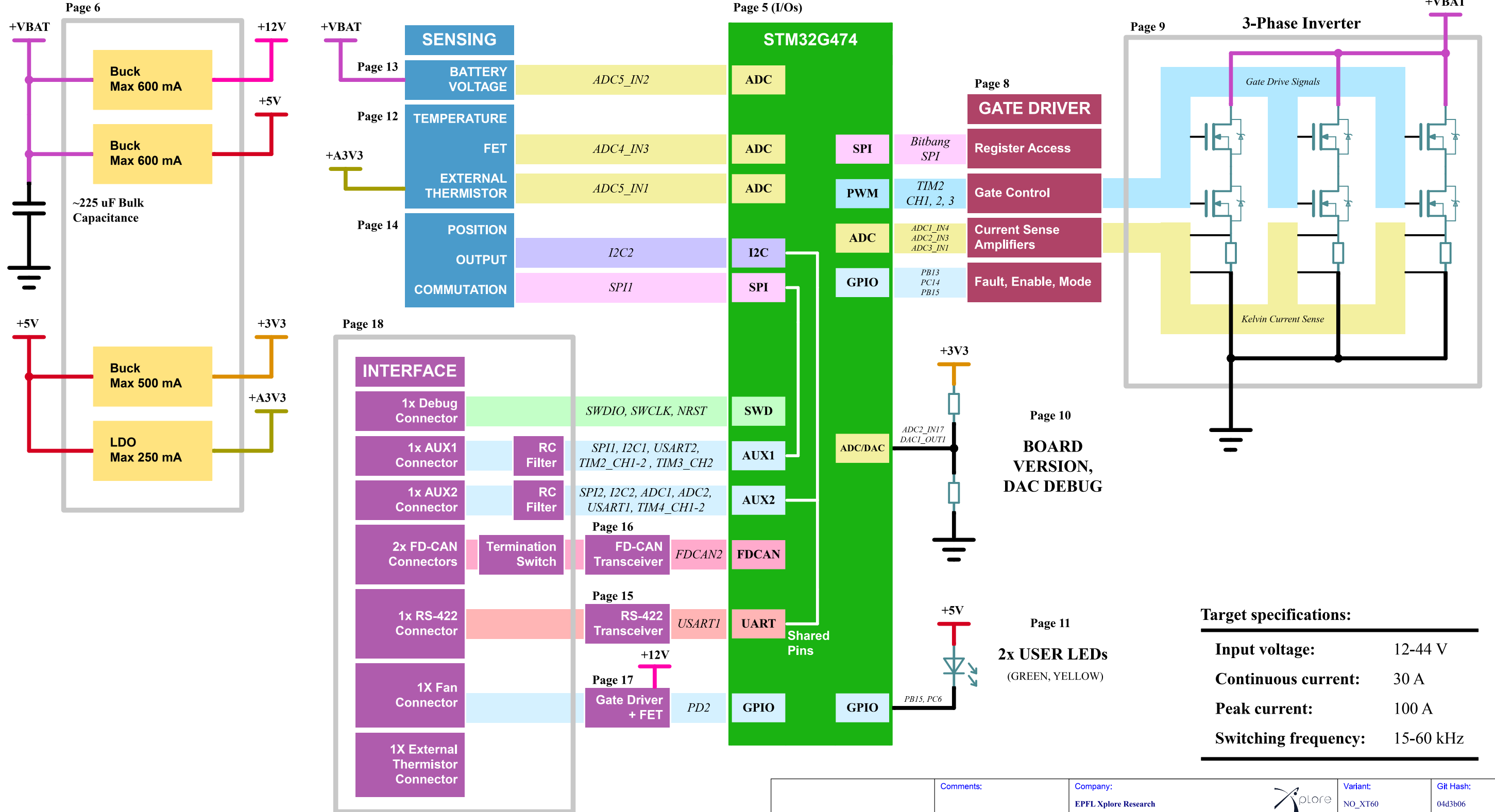
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 13-APR-2024

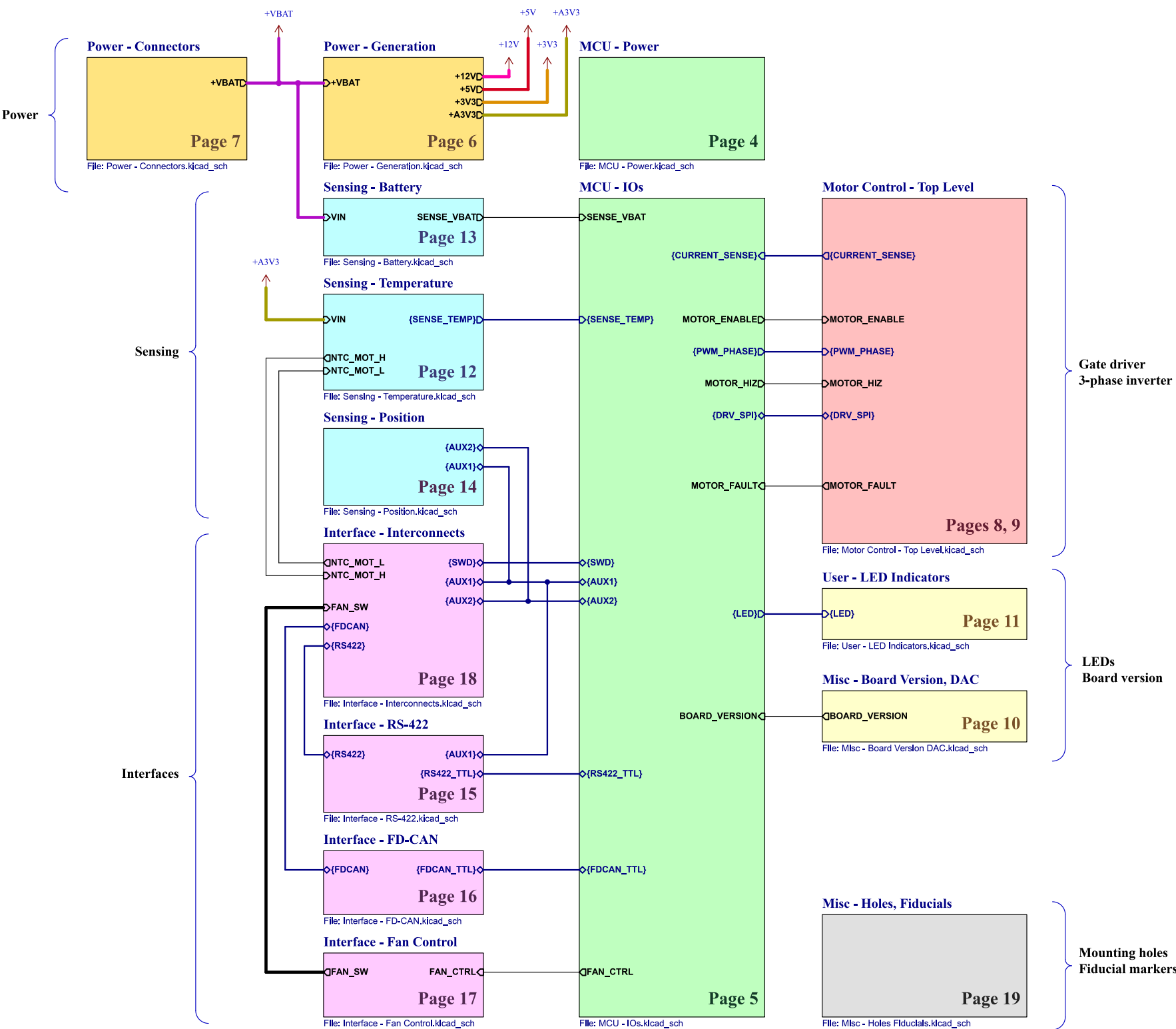
	Comments:	Company: EPFL Xplore Research			Variant: NO_XT60	Git Hash: 04d3b06
		Board Name: Amulet Motion Controller			Project Name: Chienpanzé	
	Sheet Title: Cover Page	File Name: amulet_controller.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1	
	Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 21	

[2] Block Diagram



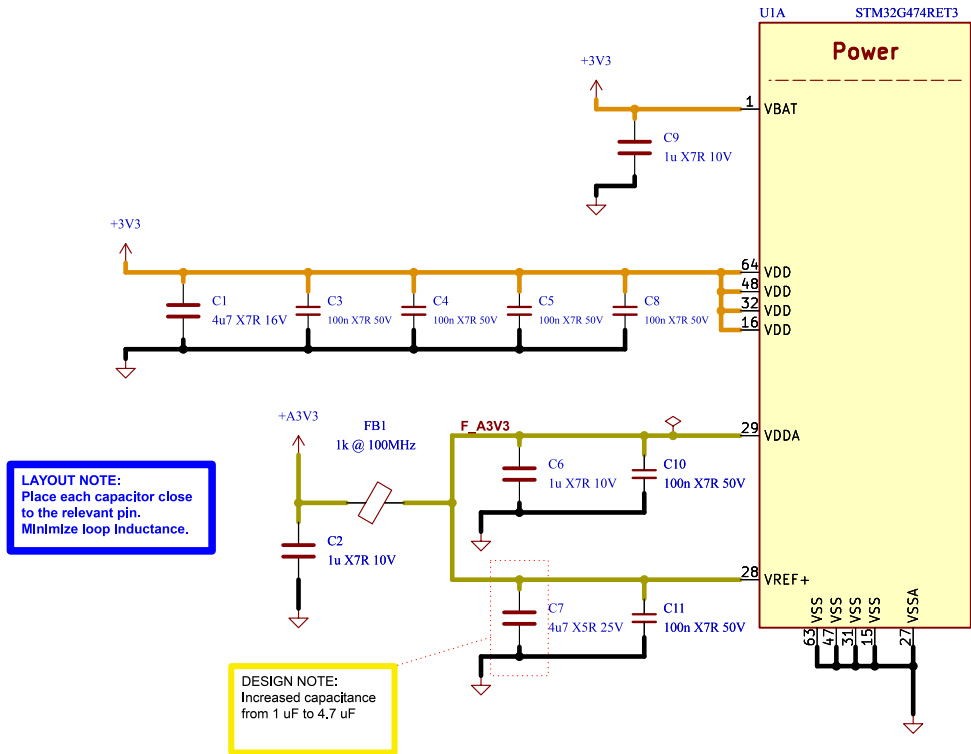
Comments:	Company: EPFL Xplore Research				Variant: NO_XT60	Git Hash: 04d3b06
	Board Name: Amulet Motion Controller			Project Name: Chienpanzé		
Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1		
Sheet Path: /Block Diagram/		Reviewer:	Size: A3	Sheet: 2 of 21		

[3] Project Architecture



Comments:	Company: EPFL Xplore Research			Variant: NO_XT60	Git Hash: 04d3b06
	Board Name: Amulet Motion Controller			Project Name: Chienpanzé	
	Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-22	Revision: 1.1
Sheet Path: /Project Architecture/		Reviewer:		Size: A3	Sheet: 3 of 21

[4] MCU - Power



Microcontroller Power Supply Scheme

	Comments: AN5346 STM32G474 Datasheet p.81 J. Pieper ADC investigation	Company: EPFL Xplore Research		Variant: NO_XT60	Git Hash: 04d3b06
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: MCU - Power	File Name: MCU - Power.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-18	Revision: 1.1
	Sheet Path: /Project Architecture/MCU - Power/		Reviewer:	Size: A4	Sheet: 4 of 21

[5] MCU - I/Os

Microcontroller I/O Domain

STM32G474RET3 U1B

I/O

SWDIO 49 PA13 SWD
SWCLK 50 PA14
NRST 7 PG10

AUX2_B 5 PF0
AUX2_A 6 PF1

FAN_CTRL 55 PD2

PC0 8 PC0
PC1 9 PC1
PC2 10 PC2
PC3 11 PC3
AUX2_C 22 PC4
LED_PWR 23 PC5
PC6 38 PC6
PC7 39 PC7
PC8 40 PC8
PC9 41 PC9
DRV_SCLK 52 PC10
DRV_MISO 53 PC11
PC12 54 PC12
DRV_MOSI 2 PC13
MOTOR_ENABLE 3 PC14
MOTOR_HIZ 4 PC15

PORT A
PORT B
PORT C
PORT D
PORT F

PA0 12 PWM_PHASEC
PA1 13 PWM_PHASEB
PA2 14 PWM_PHASEA
PA3 17 FILTER_SOA
PA4 18 BOARD_VERSION
PA5 19 AUX1_A
PA6 20 FILTER_SOB
PA7 21 AUX1_C
PA8 42 SENSE_TEMP_MOT
PA9 43 SENSE_VBAT
PA10 44 AUX2_B
PA11 45 AUX2_C
PA12 46 AUX2_I2C_PULLUP
PA15 51 AUX1_D

PB0 24 DRV_CS
PB1 25 FILTER_SOC
PB2 26 POS_SENSOR_CS
PB3 56 AUX1_E
PB4 57 AUX1_B
PB5 58 FDCAN_RX
PB6 59 FDCAN_TX
PB7 60 AUX2_D
PB8 61 AUX1_I2C_PULLUP
PB9 62 RS422_RE
PB10 63 RS422_DE
PB11 64 SENSE_TEMP_FET
PB12 65 MOTOR_FAULT
PB13 66 AUX1_A
PB14 67 LED_DBG
PB15 68

DESIGN NOTE:
We use the LQFP64 version to have extra pins for controlling an external fan.

DESIGN NOTE:
Do not forget to flash option bytes when flashing for the first time.

LAYOUT NOTE:
Footprint deviates from datasheet and IPC7351 to allow for soldermask sliver between pads according to PCBWay design rules.

SWD (Debug) Port

SWDIO
SWCLK
NRST

User LEDs

LED_DBG
LED_PWR

Connection to Gate Driver

SPI for DRV config
DRV_SCLK
DRV_MISO
DRV_MOSI
DRV_CS

Half bridge PWMs
PWM_PHASEA
PWM_PHASEB
PWM_PHASEC

Shunt outputs filtered
FILTER_SOA
FILTER_SOB
FILTER_SOC

Motor enable and fault
MOTOR_ENABLE
MOTOR_HIZ
MOTOR_FAULT

DESIGN NOTE:
PC13 is used instead of PC12 for DRV_MOSI (bitbang SPI) to be compatible with motews FW.

LAYOUT NOTE:
Shunt amplifier output low-pass filters. Place near MCU ADC pins.

Auxiliary Ports

Auxiliary port 1
AUX1_A
AUX1_B
AUX1_C
AUX1_D
AUX1_E
AUX1_I2C_PULLUP
POS_SENSOR_CS

Auxiliary port 2
AUX2_A
AUX2_B
AUX2_C
AUX2_D
AUX2_I2C_PULLUP

RS-422 port
RS422_RE
RS422_DE

5Mbps FD-CAN Bus

FDCAN_RX
FDCAN_TX

Temperature Sense

SENSE_TEMP_MOT
SENSE_TEMP_FET

Battery Voltage Sense

SENSE_VBAT

Low-Side Fan Control

FAN_CTRL

Gate Driver

MOTOR_ENABLE TP1
MOTOR_HIZ TP2
MOTOR_FAULT TP3

DRV_SCLK TP4
DRV_MISO TP5
DRV_MOSI TP6
DRV_CS TP7

PWM_PHASEA TP8
PWM_PHASEB TP9
PWM_PHASEC TP10

SOA TP11
SOB TP12
SOC TP13

Debug

SWDIO TP14
SWCLK TP15
NRST TP16

Auxiliary pins 1

AUX1_A TP17
AUX1_B TP18
AUX1_C TP19
AUX1_D TP20
AUX1_E TP21
AUX1_I2C_PULLUP TP22
POS_SENSOR_CS TP23

Auxiliary pins 2

AUX2_A TP24
AUX2_B TP25
AUX2_C TP26
AUX2_D TP27
AUX2_I2C_PULLUP TP28

RS-422

RS422_RE TP29
RS422_DE TP30

LEDs

LED_DBG TP31
LED_PWR TP32

FD-CAN

FDCAN_RX TP33
FDCAN_TX TP34

Fan

FAN_CTRL TP35

Sense

SENSE_TEMP_MOT TP36
SENSE_TEMP_FET TP37
SENSE_VBAT TP38

Test Points

TP1
TP2
TP3
TP4
TP5
TP6
TP7
TP8
TP9
TP10
TP11
TP12
TP13
TP14
TP15
TP16
TP17
TP18
TP19
TP20
TP21
TP22
TP23
TP24
TP25
TP26
TP27
TP28
TP29
TP30
TP31
TP32
TP33
TP34
TP35
TP36
TP37
TP38

Comments:
References:
Flexible I/O worked examples
Flexible I/O source configuration

Company:
EPFL Xplore Research

Board Name:
Amulet Motion Controller

Variant:
NO_XT60

Project Name:
Chienpanzé

Sheet Title:
MCU - I/Os

File Name:
MCU - I/Os.kicad_sch

Designer:
Vincent Nguyen

Date:
2023-12-20

Revision:
1.1

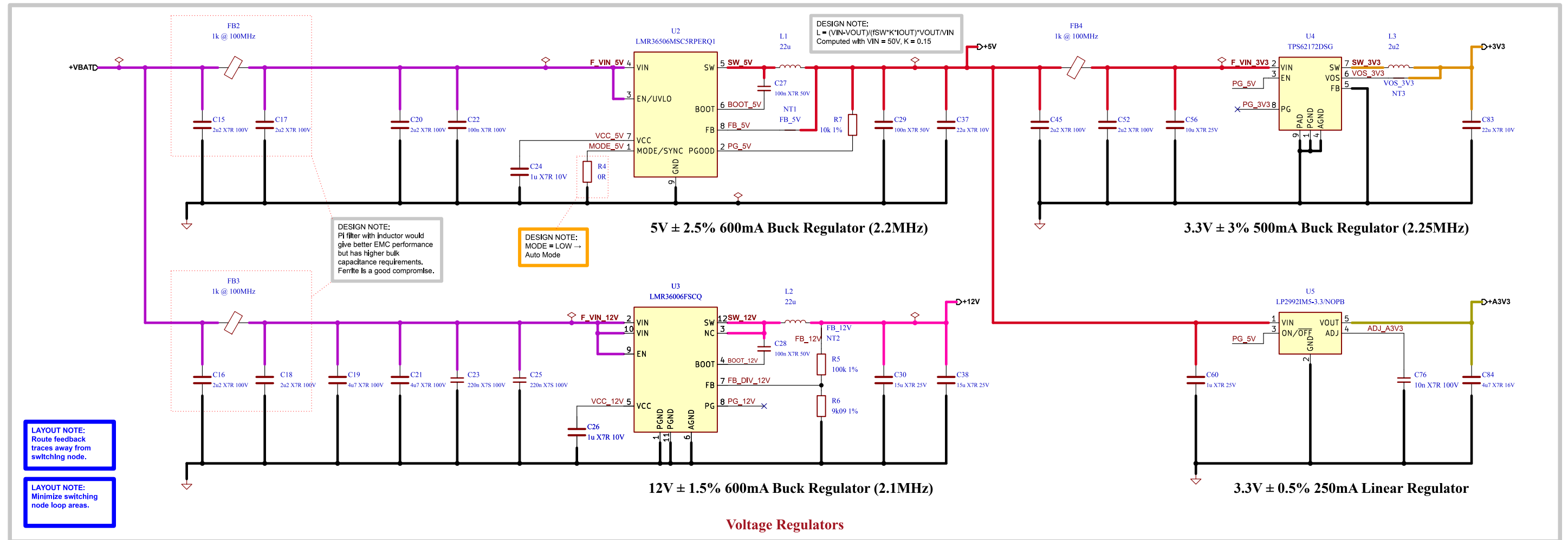
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/Project Architecture/MCU - I/Os/

Reviewer:

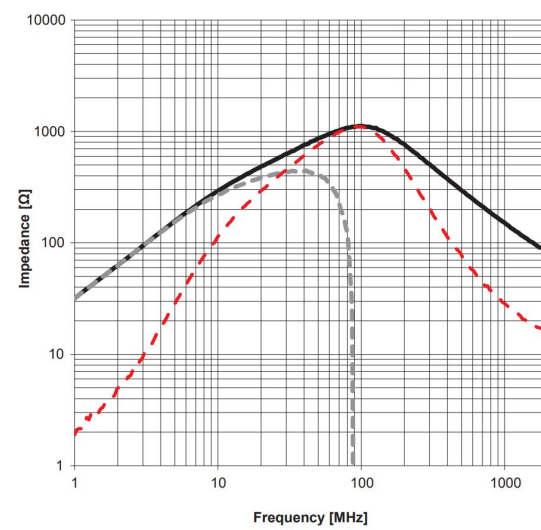
Size:
A3

Sheet:
5 of 21

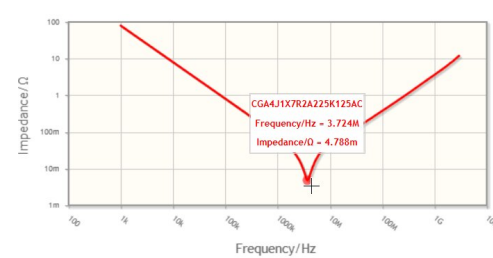
[6] Power - Generation



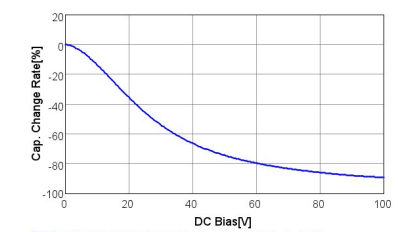
Ferrite bead Impedance vs Frequency curve



Pi filter capacitor Impedance vs Frequency curve



Bulk capacitance Derating curve



Capacitor and Ferrite Bead Parameters

Battery

+VBAT TP39 +VBAT
+VBAT TP40 +VBAT
+VBAT TP41 +VBAT

Filtered Power Rails

F_VIN_12V TP42 F_VIN_12V
F_VIN_5V TP43 F_VIN_5V
F_VIN_3V3 TP44 F_VIN_3V3

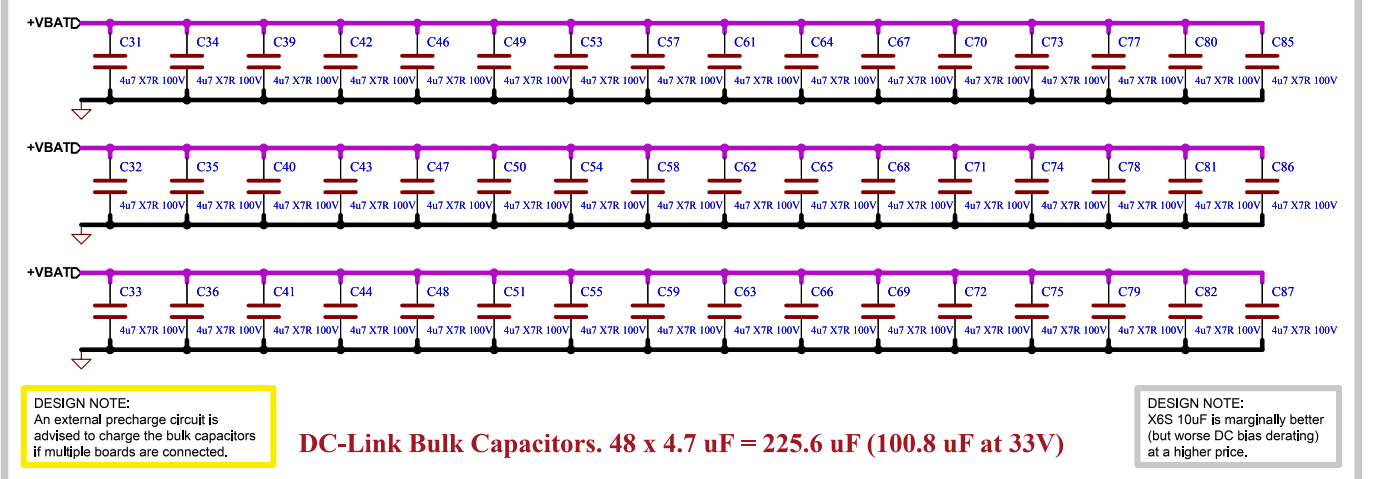
Power Rails

+12V TP45 +12V
+5V TP46 +5V
+5V TP47 +5V
+3V3 TP48 +3V3
+3V3 TP49 +3V3
+3V3 TP50 +3V3
+A3V3 TP51 +A3V3

Ground

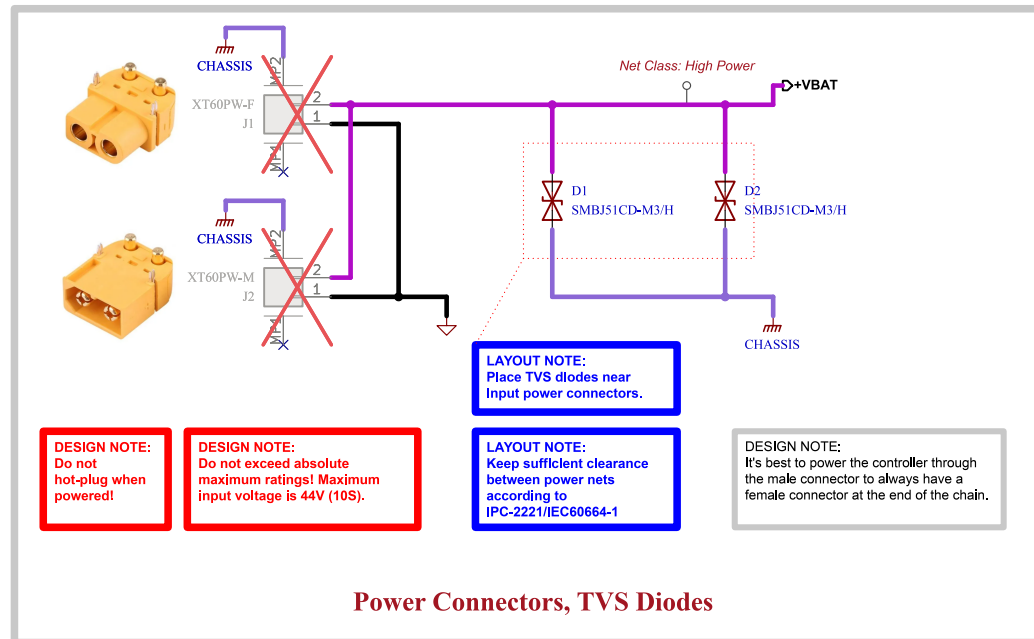
GND TP52 GND
GND TP53 GND
GND TP54 GND
GND TP55 GND
GND TP56 GND
GND TP57 GND
GND TP58 GND

Test Points



Comments: +12V (600 mA): Fan power +5V (600 mA): Gate Driver, LEDs +3V3 (500 mA): digital 3.3V +3V3A (250 mA): analog 3.3V	Company: EPFL Xplore Research		Variant: NO_XT60	Git Hash: 04d3b06
	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
Sheet Title: Power - Generation	File Name: Power - Generation.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1
Sheet Path: /Project Architecture/Power - Generation/	Reviewer:		Size: A3	Sheet: 6 of 21

[7] Power - Connectors



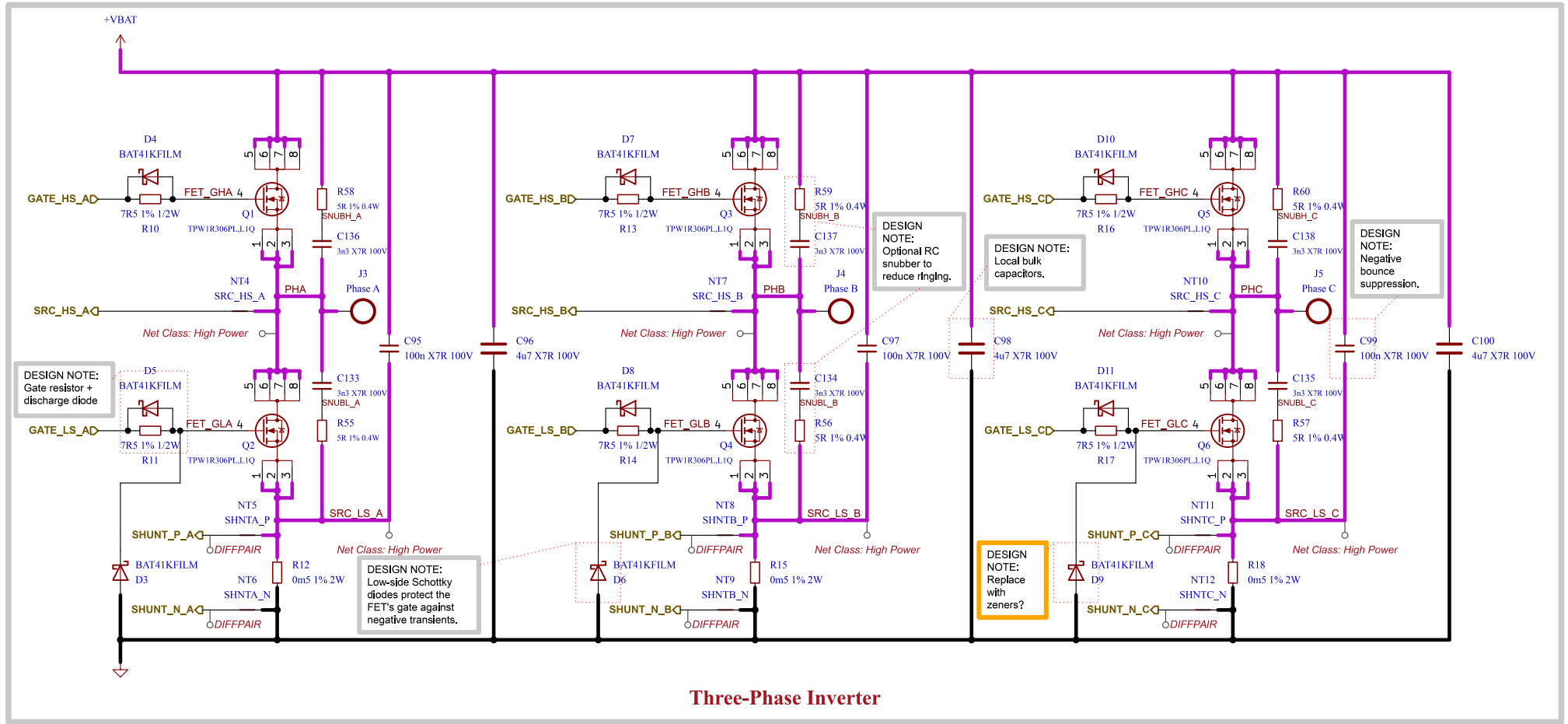
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		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Power - Connectors	File Name: Power - Connectors.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-31	Revision: 1.1
	Sheet Path: /Project Architecture/Power - Connectors/		Reviewer:	Size: A4	Sheet: 7 of 21

100-V Three-Phase Smart Gate Driver



	Comments:	Company:			Variant:	Git Hash:
		EPFL Xplore Research		NO_XT60	04d3b06	
		Board Name:			Project Name:	
		Amulet Motion Controller			Chienpanzé	
Sheet Title:	File Name:	Designer:	Date:	Revision:		
Motor Control - Top Level	Motor Control - Top Level.kicad_sch	Vincent Nguyen	2023-12-20	1.1		
Sheet Path:		Reviewer:	Size:	Sheet:		
/Project Architecture/Motor Control - Top Level/			A3	8 of 21		

[9] Motor Control - Inverter



LAYOUT NOTE:
High current traces must be carefully designed. Ensure ground return path does not cross sensitive parts of the board. Use multiple planes for higher current carrying capacity.

LAYOUT NOTE:
Keep sufficient clearance between power nets according to IPC-2221/IEC60664-1.

DESIGN NOTE:
A gate drive current that is too large can damage the FETs!

Comments:
System Design Considerations for High-Power Motor Driver Applications
Best Practices for Board Layout of Motor Drivers
Proper RC Snubber Design for Motor Drivers

Sheet Title:
Motor Control - Inverter

Sheet Path:
/Project Architecture/Motor Control - Top Level/Motor Control - Inverter/

Company:
EPFL Xplore Research

Board Name:
Amulet Motion Controller

File Name:
Motor Control - Inverter.kicad_sch

Reviewer:



Variant:
NO_XT60

Project Name:
Chienpanzé

Date:
2024-01-25

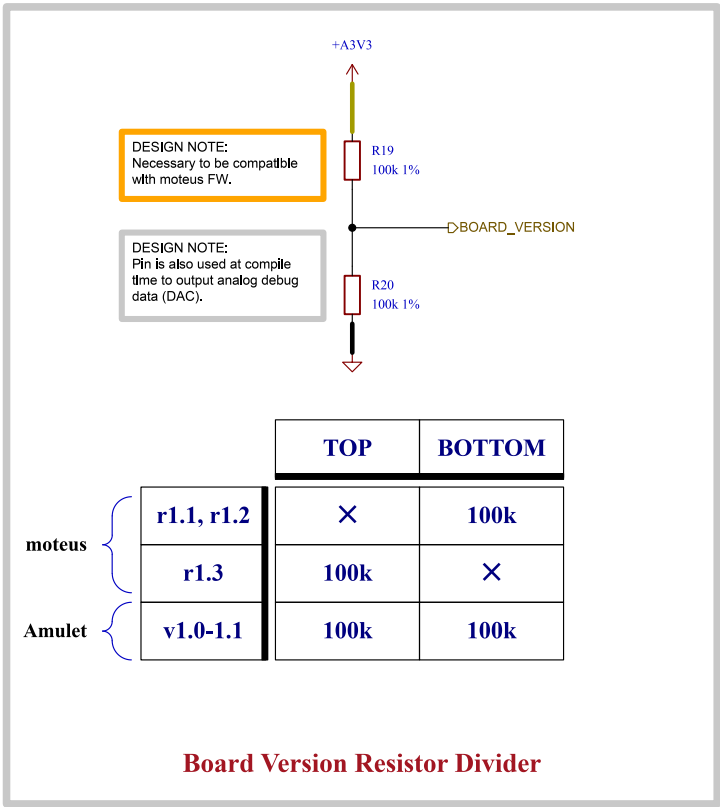
Size:
A4

Git Hash:
04d3b06

Revision:
1.1

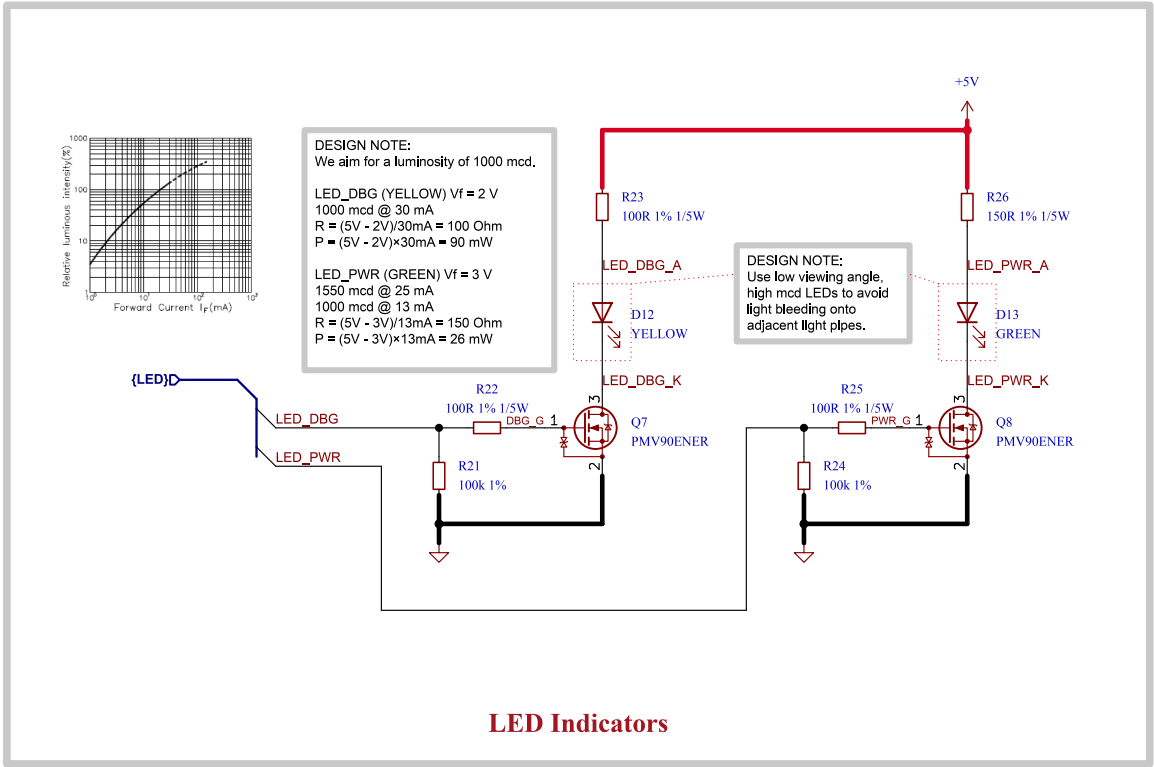
Sheet:
9 of 21

[10] Misc - Board Version, DAC



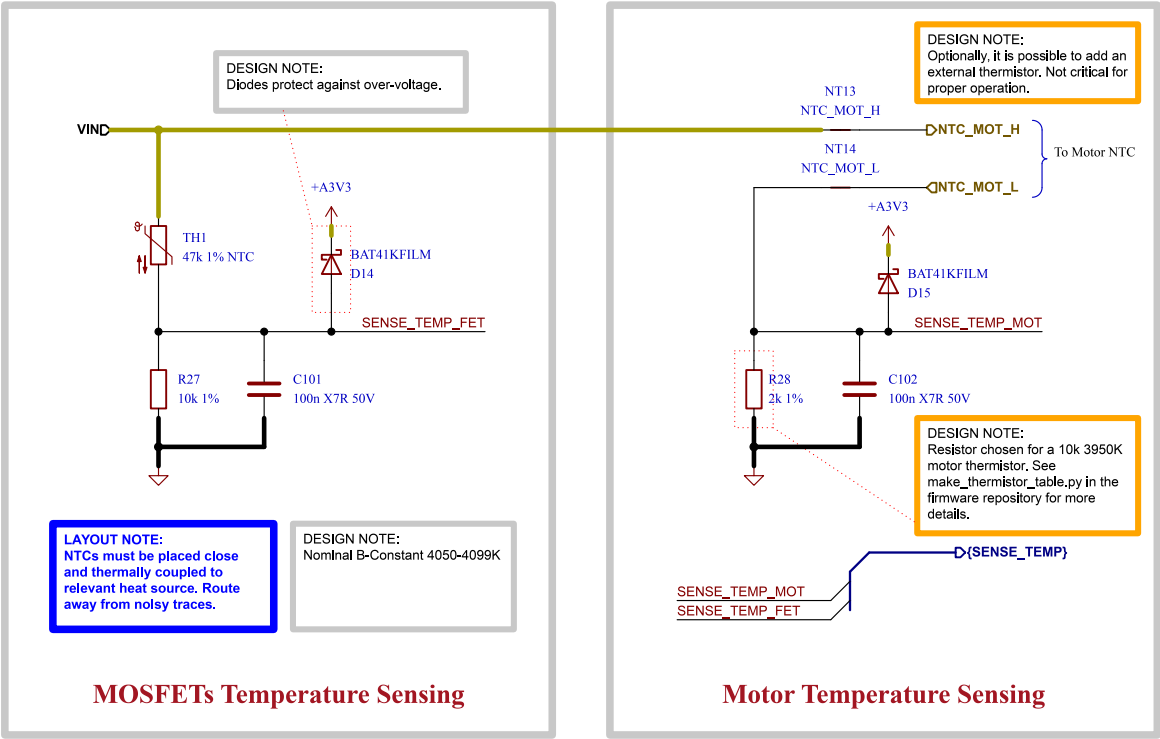
	Comments:	Company: EPFL Xplore Research		Variant: NO_XT60	Git Hash: 04d3b06
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Misc - Board Version, DAC	File Name: Misc - Board Version DAC.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1
	Sheet Path: /Project Architecture/Misc - Board Version, DAC/		Reviewer:	Size: A4	Sheet: 10 of 21

[11] User - LED Indicators



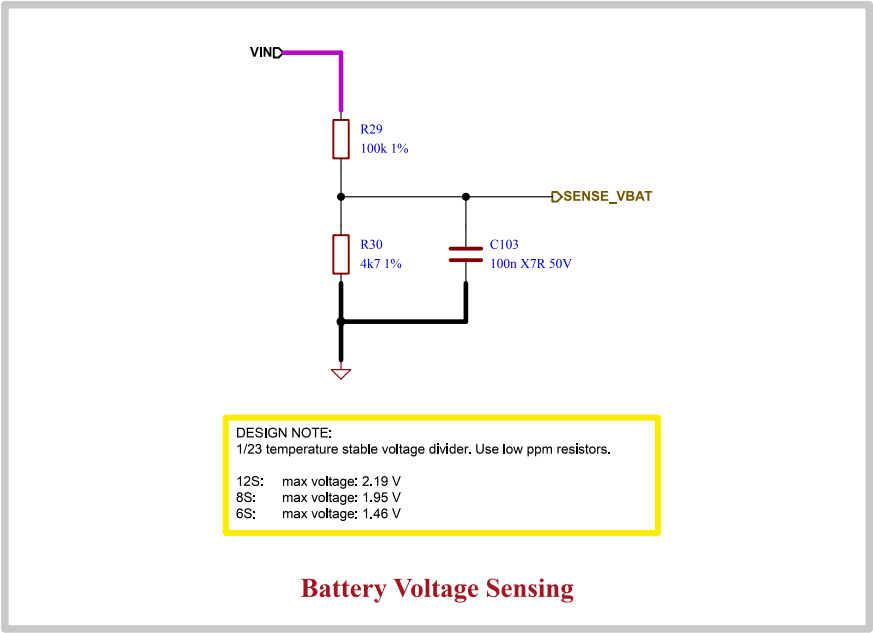
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		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: User - LED Indicators	File Name: User - LED Indicators.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-19	Revision: 1.1
	Sheet Path: /Project Architecture/User - LED Indicators/		Reviewer:	Size: A4	Sheet: 11 of 21

[12] Sensing - Temperature



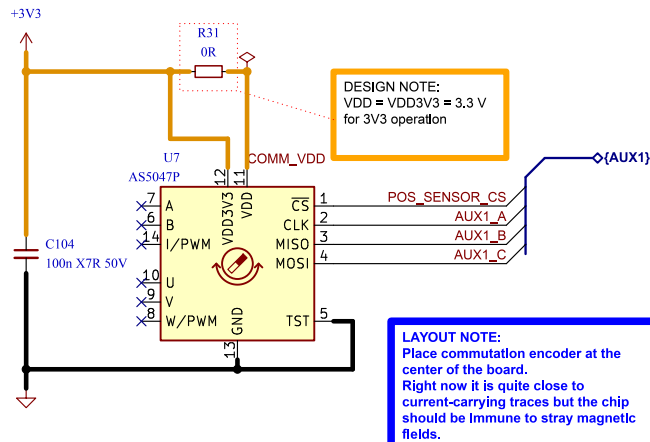
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		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Sensing - Temperature	File Name: Sensing - Temperature.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1
	Sheet Path: /Project Architecture/Sensing - Temperature/		Reviewer:	Size: A4	Sheet: 12 of 21

[13] Sensing - Battery



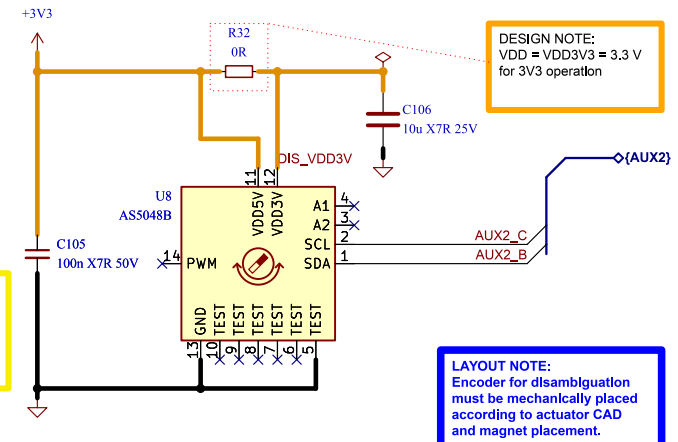
	Comments:		Company:			Variant:		Git Hash:		
			EPFL Xplore Research			NO_XT60		04d3b06		
			Board Name:				Project Name:			
			Amulet Motion Controller				Chienpanzé			
Sheet Title:		File Name:		Designer:		Date:		Revision:		
Sensing - Battery		Sensing - Battery.kicad_sch		Vincent Nguyen		2023-10-14		1.1		
Sheet Path:				Reviewer:		Size:		Sheet:		
/Project Architecture/Sensing - Battery/						A4		13 of 21		

[14] Sensing - Position



14-Bit On-Axis Magnetic Rotary Position Sensor (commutation)

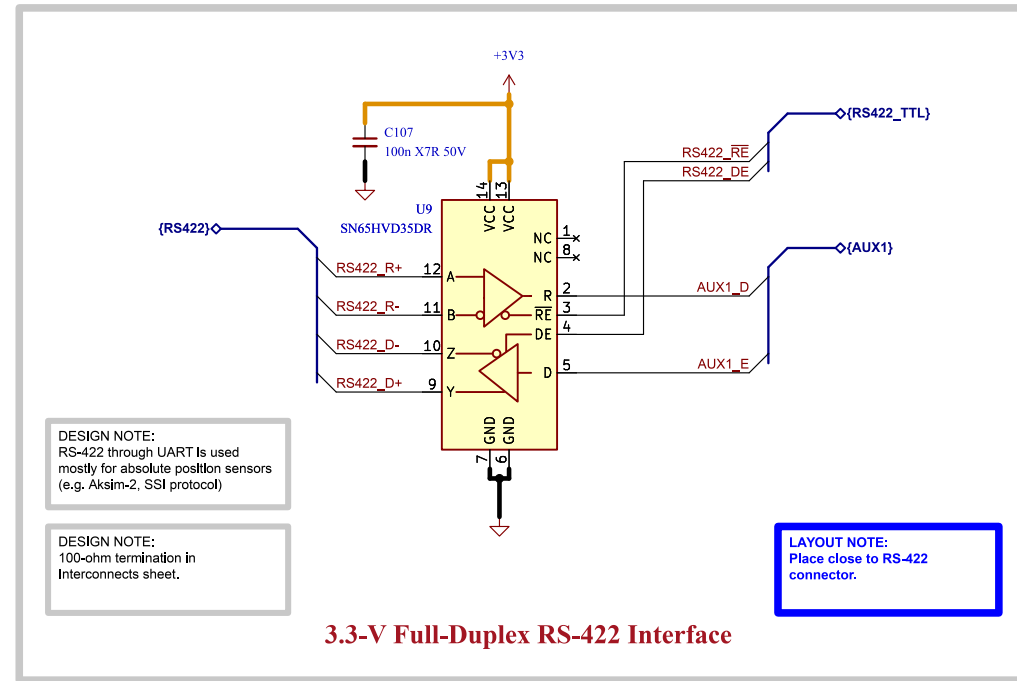
DESIGN NOTE:
AS5047P senses magnet mounted on planetary sun gear, for commutation.
AS5048B senses magnet mounted on shaft with same reduction factor as planetary gearbox for disambiguation.



14-Bit On-Axis Magnetic Rotary Position Sensor (disambiguation)

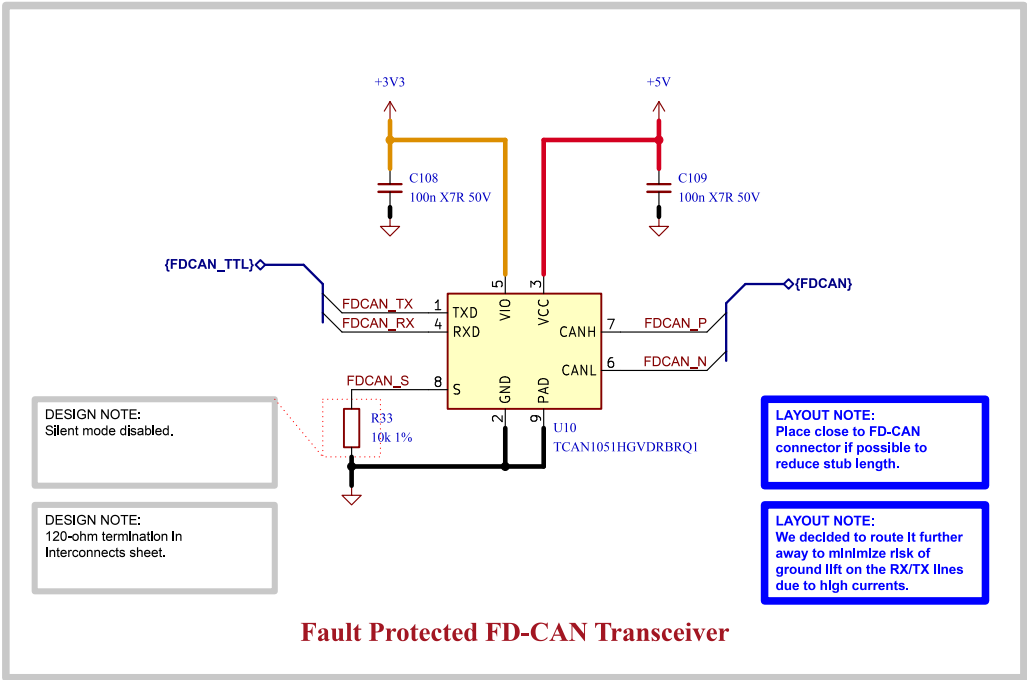
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		Board Name: Amulet Motion Controller	Project Name: Chienpanzé	
	Sheet Title: Sensing - Position	File Name: Sensing - Position.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-14
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			Sheet: 14 of 21	

[15] Interface - RS-422



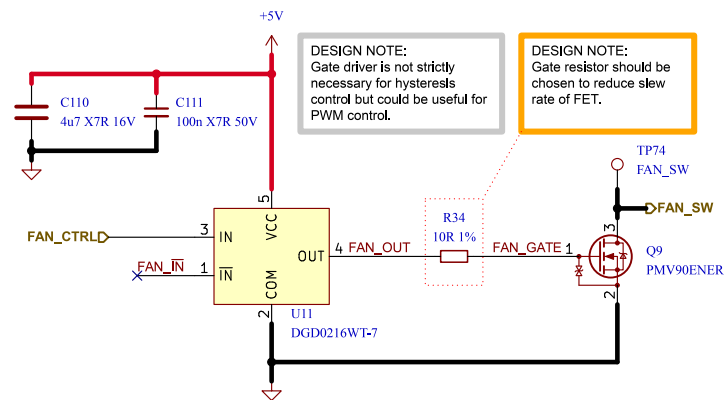
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		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Interface - RS-422	File Name: Interface - RS-422.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-15	Revision: 1.1
	Sheet Path: /Project Architecture/Interface - RS-422/	Reviewer:	Size: A4	Sheet: 15 of 21	

[16] Interface - FD-CAN



	Comments:	Company: EPFL Xplore Research		Variant: NO_XT60	Git Hash: 04d3b06
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Interface - FD-CAN	File Name: Interface - FD-CAN.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-15	Revision: 1.1
	Sheet Path: /Project Architecture/Interface - FD-CAN/		Reviewer:	Size: A4	Sheet: 16 of 21

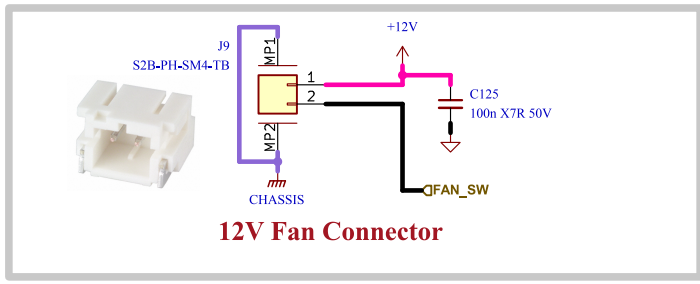
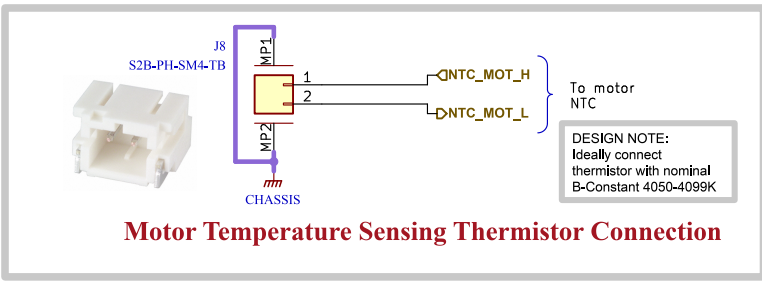
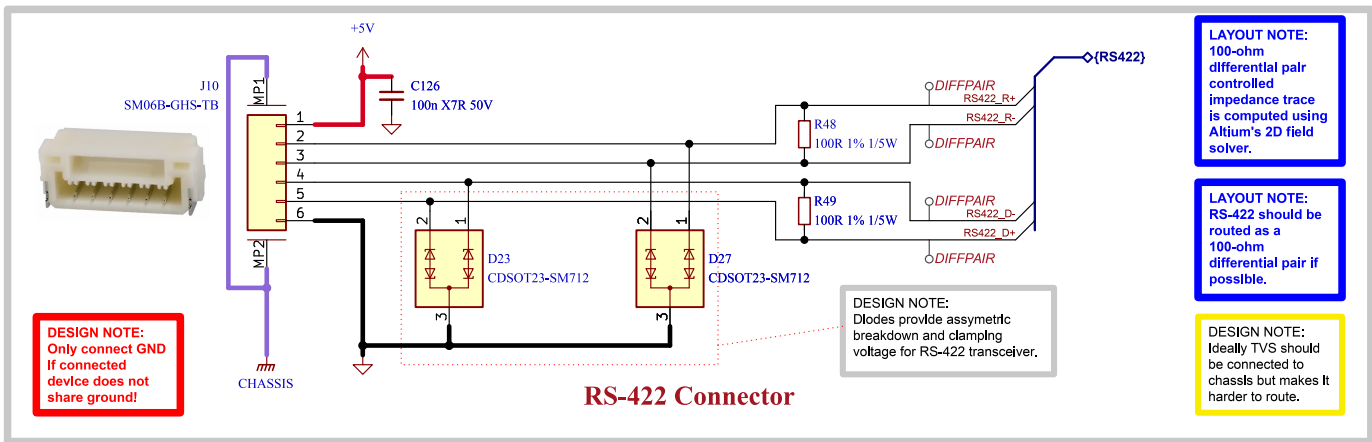
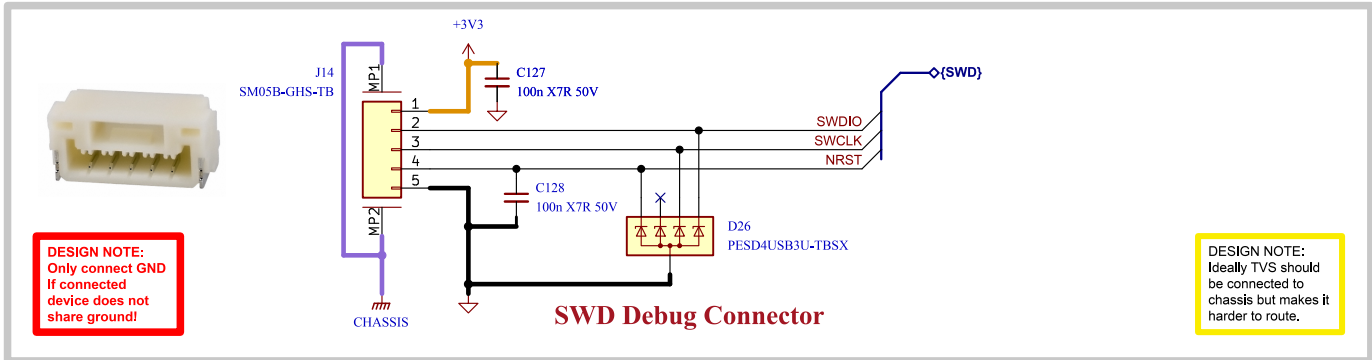
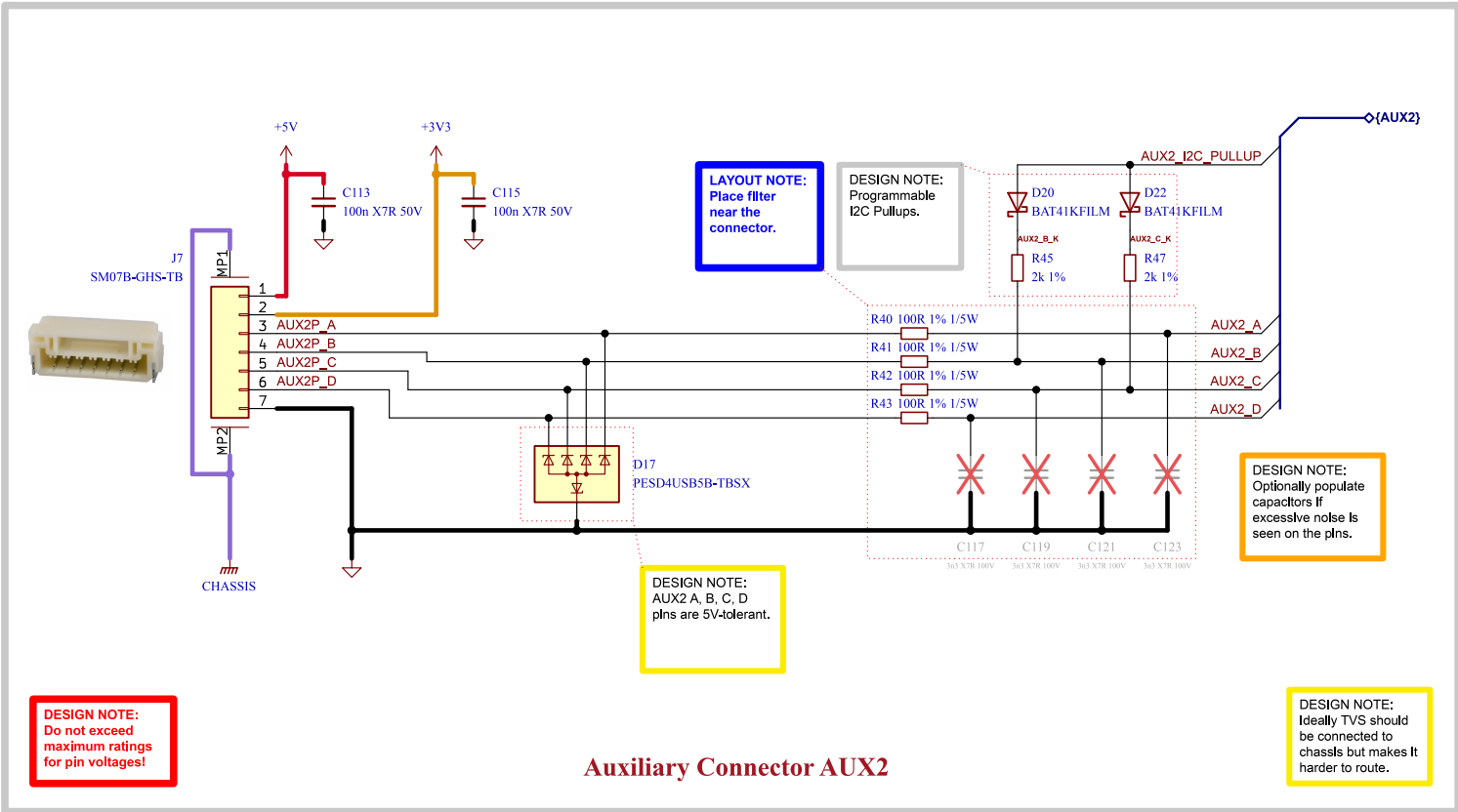
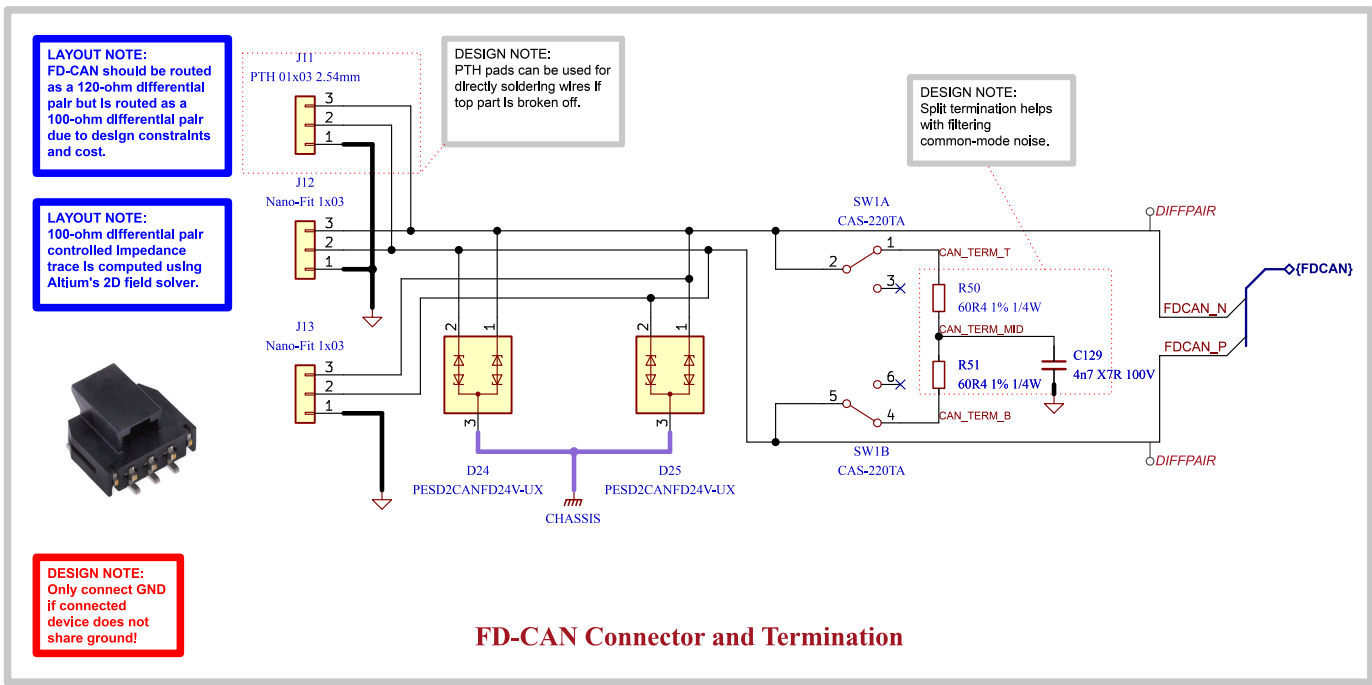
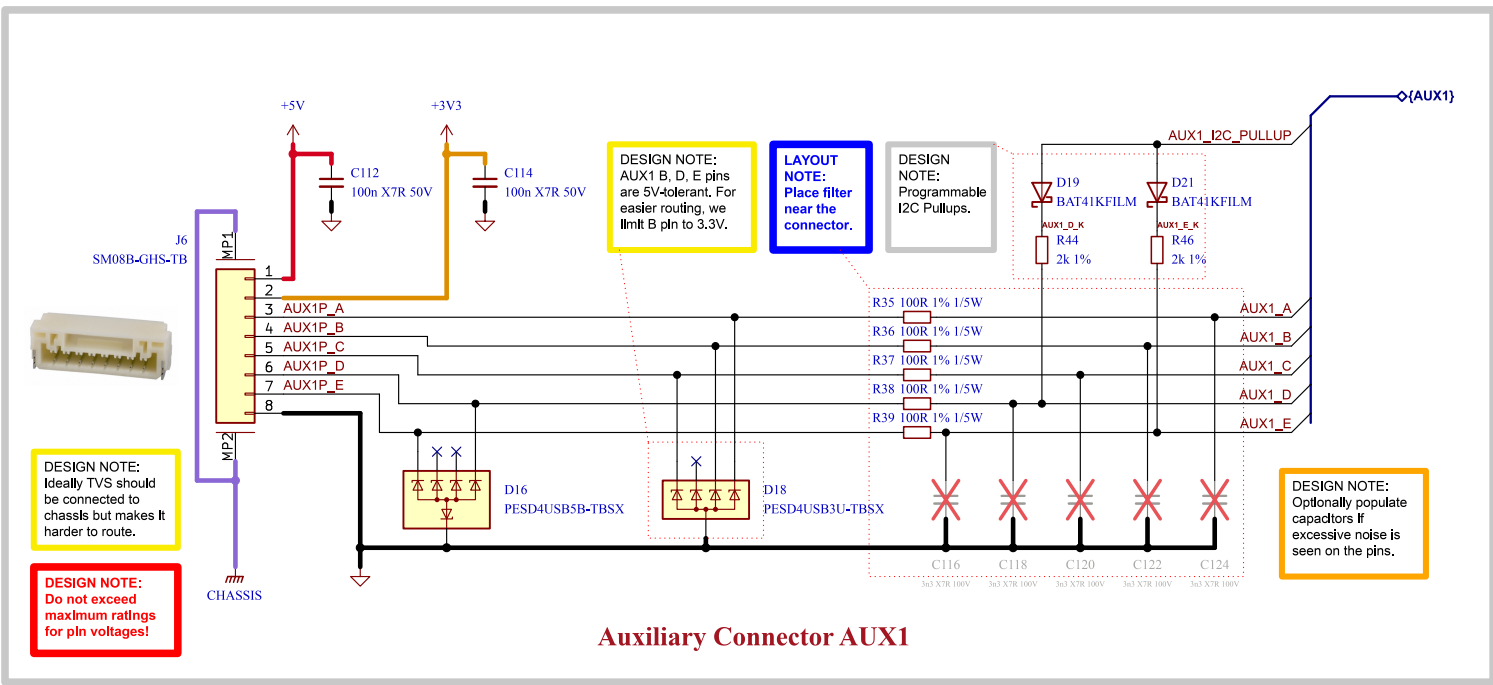
[17] Interface - Fan Control



Low-side Fan Switch and Gate Driver

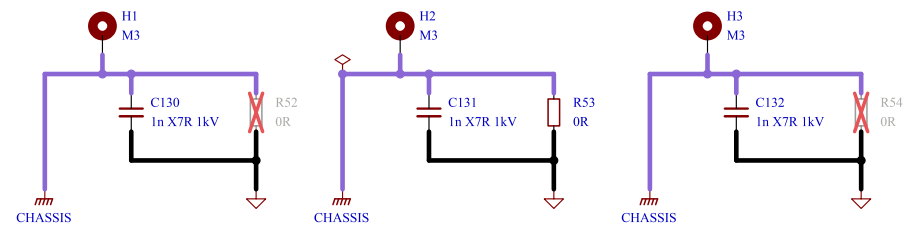
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		Board Name: Amulet Motion Controller			Project Name: Chienpanzé	
Sheet Title: Interface - Fan Control	File Name: Interface - Fan Control.kicad_sch	Designer: Vincent Nguyen		Date: 2023-11-19	Revision: 1.1	
Sheet Path: /Project Architecture/Interface - Fan Control/		Reviewer:		Size: A4	Sheet: 17 of 21	

[18] Interface - Interconnects



Comments: Reference: Flexible I/O worked examples	Company: EPFL Xplore Research		Variant: NO_XT60	Git Hash: 04d3b06
	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
Sheet Title: Interface - Interconnects	File Name: Interface - Interconnects.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-31	Revision: 1.1
Sheet Path: /Project Architecture/Interface - Interconnects/		Reviewer:	Size: A3	Sheet: 18 of 21

[19] Misc - Holes, Fiducials



DESIGN NOTE:
Optional RC filter for chassis
grounding. Capacitors offer multi-point
low-impedance connections at
high-frequencies.

DESIGN NOTE:
0-Ohm resistor provides low-impedance
bond between chassis and signal GND.
Be aware of ground loops through cable
shield.

Mounting Holes and Chassis Connection

Front



Back



Fiducial Markers

	Comments:	Company: EPFL Xplore Research		Variant: NO_XT60	Git Hash: 04d3b06
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Misc - Holes, Fiducials	File Name: Misc - Holes Fiducials.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-22	Revision: 1.1
	Sheet Path: /Project Architecture/Misc - Holes, Fiducials/		Reviewer:	Size: A4	Sheet: 19 of 21

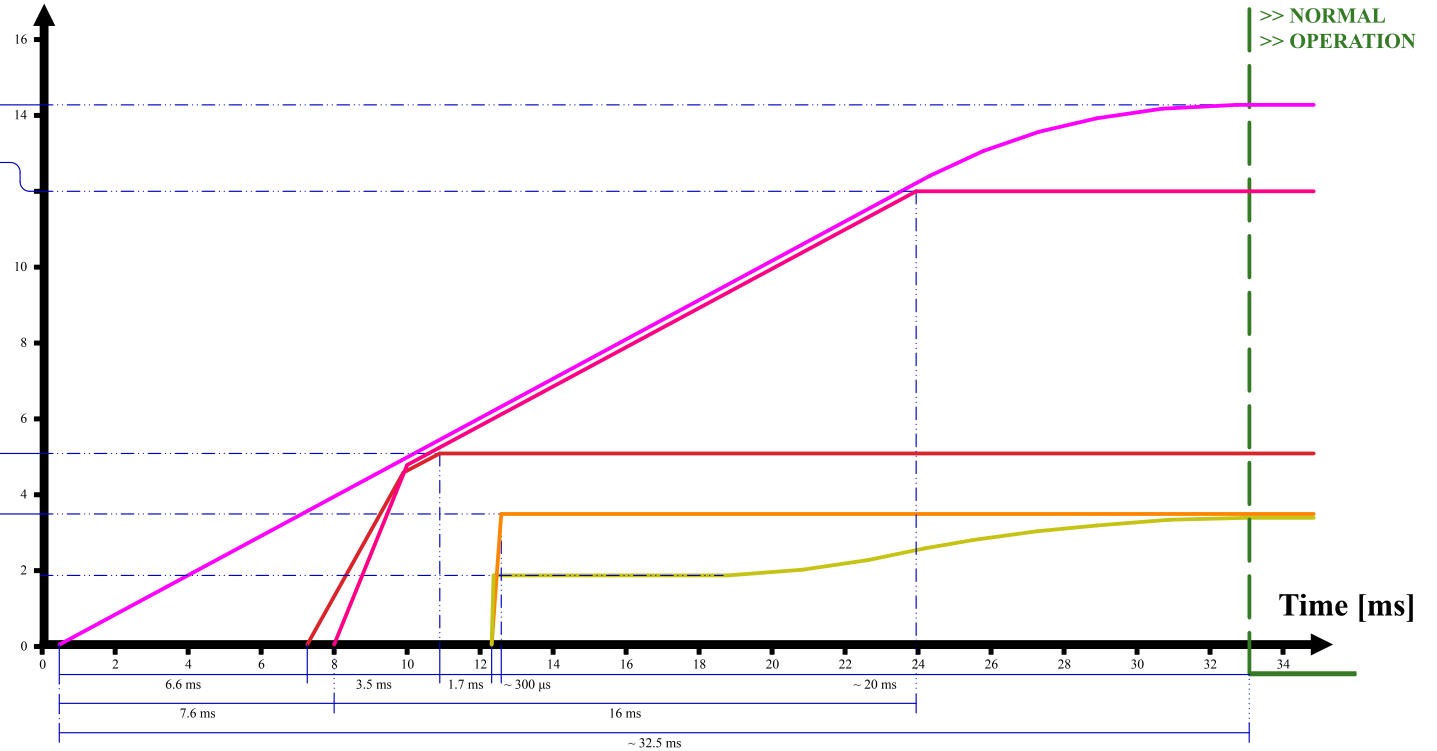
[20] Power - Sequencing

NAME	SOURCE	LEVEL
+VBAT	BATTERY	12 - 44V
+12V	LMR36006	12V $\pm$ 1.5%
+5V	LMR36506	5V $\pm$ 1.5%
+3V3	TPS62172	3.3V $\pm$ 3%
+A3V3	LP2992	3.3V $\pm$ 0.5%

DESIGN NOTE:
Power sequencing graph measured experimentally.

DESIGN NOTE:
Start-up time of 12V rail depends on
+VBAT charging time, which itself
depends on the precharge circuit.

Voltage [V]



Comments:

Company:

EPFL Xplore Research



Variant:

NO_XT60

Git Hash:

04d3b06

Board Name:

Amulet Motion Controller

Project Name:

Chienpanzé

Sheet Title:

Power - Sequencing

File Name:

Power - Sequencing.kicad_sch

Designer:

Vincent Nguyen

Date:

2024-03-12

Revision:

1.1

Sheet Path:

/Power - Sequencing/

Reviewer:

Size:

A4

Sheet:

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[21] Revision History

12-DEC-2023 - Initial Release
Variant: v1.0 Preliminary

- Changed CPH-CPL capacitor to 47nF (gate driver).
- Changed FD-CAN transceiver IC.
- Changed FETs for top cooled variant.
- Added TVS protection and termination switch to FD-CAN.
- Added low-side switched 12V 600mA source for external fan.
- Added LDO for analog supply.
- Changed input power TVS diode to bidirectional and added one diode per connector.
- Moved SOx low-pass filter to MCU section. Should be placed near MCU to avoid noise coupling into ADC lines.
- Added second onboard I2C magnetic encoder for disambiguation.
- Switched PWM_PHASEA with PWM_PHASEC on STM32G474 pinout for easier routing.
- Changed RS422 pinout on connector.
- Added ESD protection to all interfaces.
- Added overvoltage protection on thermistor ADC inputs.
- Changed buck regulators to optimize for low noise.
- Added Pi filters to inputs of buck regulators and MCU analog supply.
- Added decoupling caps next to power pins of connectors.

25-JAN-2024 - First Revision
Variant: v1.0 Checked

- Added controller target specifications.
- Replaced 5V 300mA buck converter with 600mA version.
- Added credits to moteus on cover page.
- Added optional RC-Snubber to power stage.
- Increased chassis length to go around the board.
- CAN and power TVS diodes now go to chassis.
- Changed clearance between nets to respect IEC60664-1 where possible.
- Rectified comment on precharge.
- Changed power TVS diode reference designator from "U" to "D".
- Replaced chassis-GND capacitor by 1nF 1kV.

12-MAR-2024 - First Revision
Variant: v1.0 Released

- Modified power sequencing graph according to experimental data.

13-APR-2024 - Second Revision
Variant: v1.1 Released

- Added RC snubber passive values.
- Added more vias for VBUS and LMR36006 GND pads.
- Changed board version voltage reference from +3V3 to +A3V3.
- Changed motor thermistor resistor divider to 2kOhm for a 10k 3950K thermistor.

Comments:	Company: EPFL Xplore Research				Variant: NO_XT60	Git Hash: 04d3b06
	Board Name: Amulet Motion Controller			Project Name: Chienpanzé		
Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Vincent Nguyen	Date: 2024-01-03	Revision: 1.1		
Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 21 of 21		